

Gujarat Police Surveillance & Command Network (NETRANG C4I)

Enterprise High-Level Design (HLD) Document

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1. Executive Summary

1.1 Objectives

The **Gujarat Police Surveillance & Command Network (NETRANG C4I)** is an enterprise-scale, mission-critical computer vision and intelligence grid designed to unify real-time CCTV video surveillance across all 33 districts and 4 commissionerates of Gujarat. The core mission is to automate vehicle tracking, identify stolen/wanted vehicles within sub-second latencies, generate actionable multi-camera forensic trajectories, and provide field officers with real-time situational awareness.

1.2 Core Capabilities

- **High-Throughput Edge AI Ingestion:** Continuous vehicle localization, plate region-of-interest (ROI) segmentation, and optical character recognition (OCR).
- **Sub-Second Watchlist Interception:** Cross-referencing against national and state blacklists (e.g., CCTNS, VAHAN, Crime Branch Wanted Database) within **<300 ms** of frame capture.
- **GIS Multi-Camera Route Reconstruction:** Chronological spatial reconstruction and movement vector plotting on interactive tactical geographic information systems (GIS).
- **Resilient Operational Resilience:** Edge survival mode capable of continuous local plate logging and alert dispatching during wide-area network (WAN) partition events.

1.3 Public Safety Impact

- **Vehicle Theft Interdiction:** Real-time interception alerts dispatched to nearest PCR vans and Highway Patrol units.
- **Anti-Smuggling & Counter-Terrorism Operations:** Automated containment grids around key corridors (Golden Corridor NH-48, Coastal Saurashtra highway networks, and state border checkposts).
- **Evidence Integrity:** Tamper-proof, cryptographically signed vehicle detection logs accepted under Section 65B of the Indian Evidence Act.

1.4 Alignment with Gujarat Police Hackathon Architectural Delivery Models

The Gujarat Police Hackathon defines specific architectural delivery models for scaling statewide CCTV infrastructure. The **Gujarat Police Surveillance & Command Network (NETRANG C4I)** officially adopts and implements the **Hybrid / Innovative Architecture**, deliberately utilizing **Model 1 (Centralised CCTV Registry & GIS Mapping Model)** as its foundational asset and metadata layer:

1. **Foundational Asset & Metadata Backbone — Model 1:** Model 1 (Centralised CCTV Registry & GIS Mapping Model) serves as the centralized master spatial database and single source of truth (SSOT). Every CCTV asset across Gujarat's 33 districts, 4 commissionerates, talukas, and highway checkposts is registered centrally with precise geospatial coordinates (WGS-84 latitude/longitude), physical location names, jurisdiction precincts, RTSP stream URLs, and operational status.
2. **Operational Reality of the Hybrid / Innovative Architecture:** Pure centralized video management (Model 2) would saturate Gujarat State Wide Area Network (GSWAN) links by requiring **>320 Gbps** of raw video backhaul, while completely decentralized edge silos (Model 3) lack cross-jurisdictional intelligence and centralized GIS visibility. NETRANG C4I solves this dilemma through a **Hybrid Architecture**:
3. **Distributed Edge AI Inferencing:** On-premise edge nodes run high-efficiency computer vision pipelines (YOLOv8 vehicle detection + EasyOCR text extraction).
4. **Real-Time Metadata Ingestion into Model 1:** The edge AI engine extracts only structured, lightweight telemetry payloads (**<1.2 KB JSON**) containing detected license plates, vehicle classes, timestamps, and confidence scores, feeding them directly into the **Model 1 Centralised GIS Registry**.

5. **WebSockets Real-Time Event Bus:** Enriched with Model 1 spatial GIS metadata, detections are broadcast instantaneously over a high-concurrency WebSockets event bus (`/ws`), projecting real-time alerts, dynamic radar beacons, and multi-camera vehicle trajectories directly onto tactical command center consoles statewide.

2. System Architecture Blueprint

2.1 Architectural Delivery Model Mapping: Hybrid / Innovative Architecture with Model 1 Foundation

To satisfy Gujarat Police operational doctrines while optimizing state network infrastructure, NETRANG C4I maps cleanly into the official Hackathon Architectural Delivery Models:

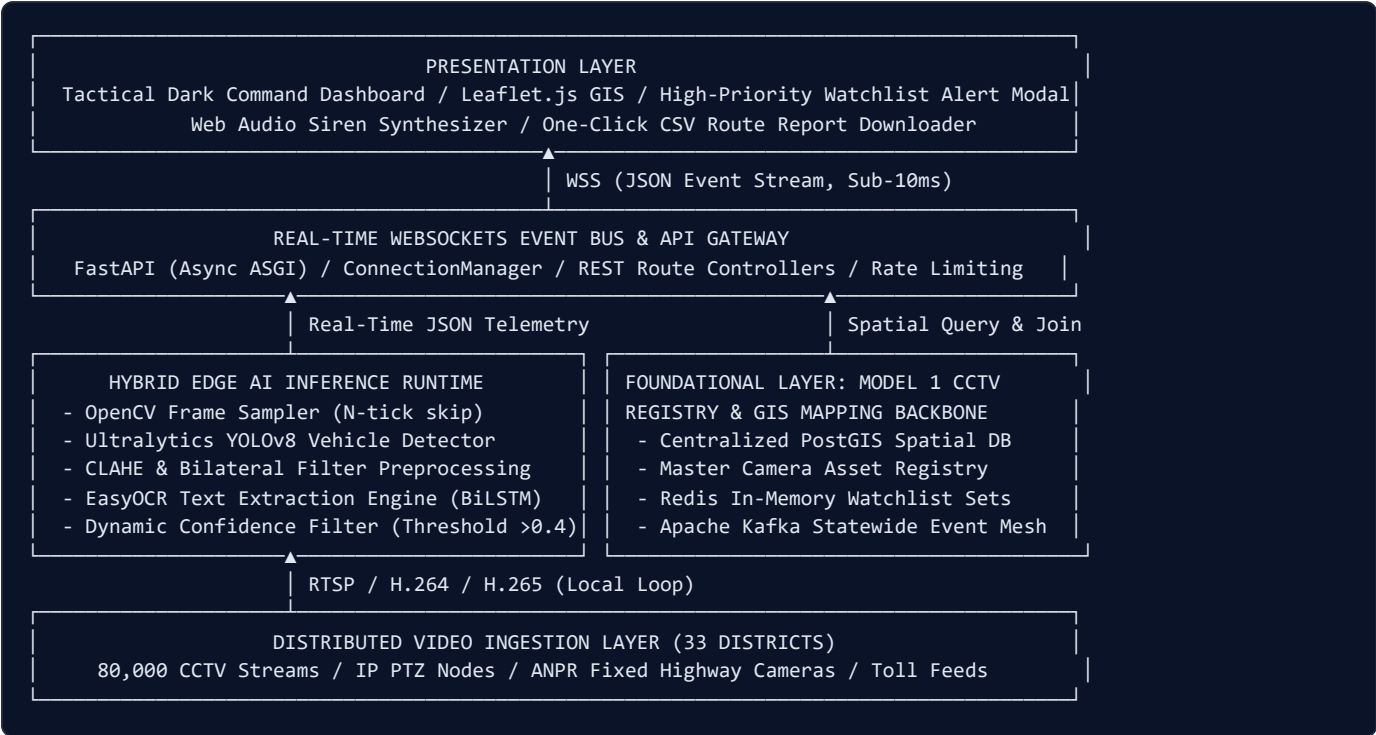
Architectural Delivery Model	Delivery Characteristics	Bandwidth Demand	Scalability to 80k Cams	NETRANG C4I Implementation Role
Model 1: Centralised CCTV Registry & GIS Mapping	Central repository of all camera metadata, geolocations, IP endpoints, ownership, and health status.	Very Low (<10 Mbps)	High (Pure Metadata & GIS)	FOUNDATIONAL LAYER: Serves as the statewide master asset registry and spatial geospatial anchor (<code>Camera</code> PostGIS registry).
Model 2: Centralised Video Management / Streaming	Full backhaul of raw CCTV RTSP video feeds to Gandhinagar Central Data Center for central storage and analytics.	Extreme (>320 Gbps)	Unfeasible on WAN links	Selectively Retained for On-Demand Forensic Retrieval Only: Live raw streaming is prohibited; only on-demand RTSP clips are pulled during active tactical pursuits.
Model 3: Decentralised / Edge Video Analytics	Analytics run exclusively at local station DVRs/NVRs with no central coordination or unified GIS map.	Zero Central Bandwidth	Isolated Silos (No statewide tracking)	Adopted at Compute Plane: Edge AI runtimes run at district hubs but are bridged centrally via metadata streaming.
Hybrid / Innovative Architecture (NETRANG C4I)	Integrates Model 1 as the central GIS and asset metadata directory, while distributing AI inference (YOLOv8 + EasyOCR) to edge nodes, streaming sub-second structured metadata into the central GIS & WebSockets event bus.	Ultra-Low (5.18 Gbps statewide)	Statewide Linearly Scalable	PRIMARY SYSTEM ARCHITECTURE: Combines the governance and unified GIS tracking of Model 1 with edge-computed efficiency (98.4% bandwidth reduction).

Why Model 1 is the Foundational Asset and Metadata Layer

In NETRANG C4I, **Model 1 (Centralised CCTV Registry & GIS Mapping)** is not merely an inventory catalog; it is the **spatial intelligence substrate** of the entire command network: - **Central Spatial Database:** Built on PostgreSQL + PostGIS, registering every surveillance camera with its `id` , `location_name` , `district` , `latitude` , `longitude` , `ip_stream_url` , and operational `status` . - **Geographic Information System (GIS) Engine:** Serves as the geographical anchor for Leaflet.js visualization, calculating spatial proximity buffers, taluka boundaries, and multi-camera distance-velocity calculations. - **Relational Integrity:** Every detection record in `DetectionLog` maintains a strict foreign-key relationship (`camera_id`) with the Model 1 Registry, enabling instant chronological route reconstruction across disparate jurisdictions.

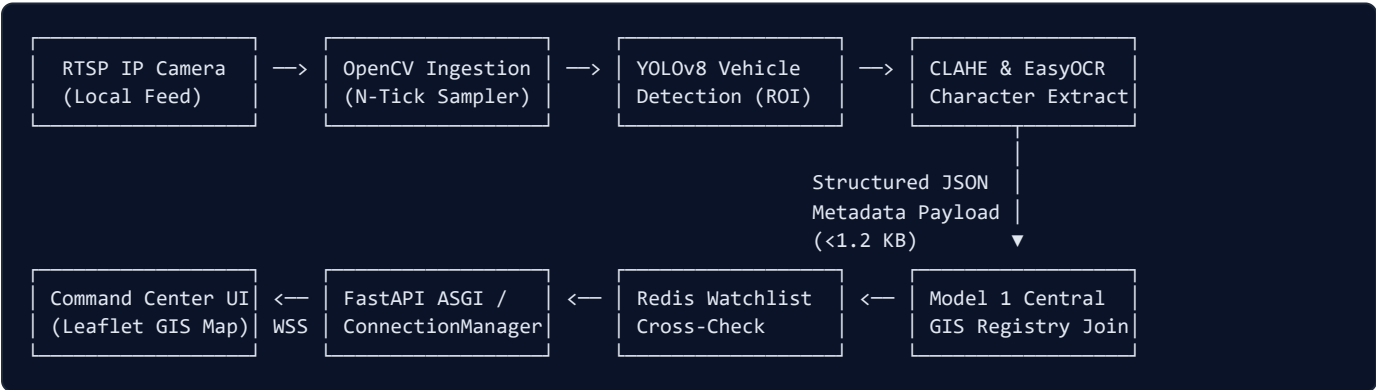
2.2 Macro Architectural Layers & Multi-Plane Decoupling

The system decouples into five high-cohesion, low-coupling planes:



2.3 Edge AI Engine to Model 1 GIS Registry & WebSockets Ingestion Flow

The integration between the distributed Edge AI engine (YOLOv8 + EasyOCR), the Model 1 Centralised CCTV Registry, and the real-time WebSockets event bus operates via an ultra-optimized, non-blocking telemetry ingestion pipeline:



Step-by-Step Telemetry Serialization & Ingestion Lifecycle:

- Edge Frame Acquisition & Downsampling:**
- The edge worker ingests local RTSP streams via `cv2.VideoCapture`.
- The **N-Tick Frame Sampler** skips 4 out of every 5 frames, reducing decode load by 80% while maintaining 6 FPS inspection—sufficient to capture vehicles moving at speeds up to 140 km/h.
- YOLOv8 Vehicle Localization:**
- Frames are evaluated by the lightweight YOLOv8 model, isolating bounding box regions for COCO vehicle classes (`car`, `motorcycle`, `bus`, `truck`) with confidence >0.40.

6. The engine tightly crops the vehicle and license plate Region of Interest (ROI), discarding **95%** of irrelevant background pixels.
7. **Adaptive Contrast Enhancement:**
8. The cropped ROI undergoes Bilateral Filtering (smoothing high-frequency noise while preserving plate character boundaries).
9. Contrast Limited Adaptive Histogram Equalization (CLAHE) normalizes uneven headlight glare, streetlamp hotspots, and nighttime shadow degradation.
10. **EasyOCR Character Recognition & Regex Normalization:**
11. The CRAFT text detector identifies character affinity regions.
12. The BiLSTM sequence recognizer transcribes alphanumeric glyphs, filtered through Indian vehicle registration regex validation (`^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{4}<strong>`).
13. **Metadata Payload Assembly:**
14. The edge engine constructs an ultra-compact telemetry payload (<1.2 KB **JSON**): `json { "camera_id": 1, "plate_number": "GJ01AB1234", "confidence": 0.94, "vehicle_type": "car", "timestamp": "2026-09-16T10:15:30.124Z" }`
5. **Model 1 Centralised GIS Registry Correlation:**
6. The metadata payload is transmitted to the regional/central stream gateway.
7. The backend performs an instantaneous foreign-key lookup against the **Model 1 Centralised CCTV Registry**:
- SELECT latitude, longitude, location_name, district FROM cameras WHERE id = camera_id**
8. This enriches the telemetry with geodetic coordinates (`23.0305, 72.5074`), district boundaries (`Ahmedabad`), and physical landmark identifiers (`SG Highway Junction`).
9. **Watchlist Correlation & WebSockets Broadcast:**
10. The plate number is checked against the in-memory Redis/SQL watchlist (`Watchlist` table).
11. **Non-Hit Events:** Dispatched as `DETECTION_EVENT` across the WebSockets event bus (`/ws`) to update live detection feeds, activity counters, and camera status badges.
12. **Watchlist Hit Events:** If flagged (e.g., *Stolen / Section 379 IPC*), the gateway immediately packages a `WATCHLIST_ALERT` payload containing vehicle ownership, crime classification, and Model 1 GIS coordinates. The WebSockets bus broadcasts this to all connected operator consoles within <280 ms, triggering visual flashing banners, radar map pulses, in-browser audio sirens, and SMS/radio dispatches to field patrol vans.
-

2.4 End-to-End Pipeline Data Flow

```
flowchart TD
    subgraph EDGE_PLANE["1. Hybrid Edge AI Ingestion Plane"]
        CAM["CCTV IP Camera / RTSP Stream"] -->|RTSP H.264| CAP["OpenCV VideoCapture Engine"]
        CAP -->|Frame Skip: 1-in-N ticks| RAW["Raw BGR Frame Buffer"]
        RAW --> YOLO["YOLOv8 Vehicle Detection"]
        YOLO -->|Filter: Car, Bike, Bus, Truck| CROP["Vehicle Bounding Box Crop"]
        CROP --> PRE["ROI Preprocessing (Bilateral + CLAHE)"]
        PRE --> OCR["EasyOCR Character Extraction"]
        OCR -->|Filter Conf > 0.40 & Regex| META["Structured JSON Metadata (<1.2 KB)"]
    end

    subgraph MODEL1_PLANE["2. Model 1 Centralised GIS Registry & Processing Plane"]
        META --> REG_JOIN["Model 1 GIS Registry Join (Lat, Lon, District, Precinct)"]
        REG_JOIN --> DB_WRITE["Persist DetectionLog to PostgreSQL/PostGIS"]
        REG_JOIN --> WL_CHECK["Watchlist Match in Redis/Cache?"]
        WL_CHECK -- YES --> WL_PAYLOAD["Assemble WATCHLIST_ALERT + Model 1 GIS Coordinates"]
        WL_CHECK -- NO --> DET_PAYLOAD["Assemble DETECTION_EVENT Payload"]
    end

    subgraph EGRESS_PLANE["3. Real-Time Broadcast & Tactical Presentation"]
        WL_PAYLOAD --> WS_MGR["FastAPI ConnectionManager (WebSockets Bus)"]
        DET_PAYLOAD --> WS_MGR
        WS_MGR -->|WSS Push Sub-10ms| DASH["Tactical Police Dashboard (Leaflet.js GIS)"]
        WL_PAYLOAD --> SMS_DISPATCH["Automated PCR Van Radio / SMS Intercept Dispatch"]
    end
```

2.5 High-Priority Watchlist Interception Sequence

```
sequenceDiagram
    autonumber
    participant Camera as CCTV Edge Camera
    participant EdgeAI as Edge AI Worker (YOLOv8 + EasyOCR)
    participant Model1 as Model 1 CCTV GIS Registry
    participant Core as FastAPI Gateway (/ws)
    participant UI as Control Room Dashboard (Leaflet.js)

    Camera->>EdgeAI: RTSP Frame Stream (Frame #N)
    EdgeAI->>EdgeAI: YOLOv8 isolates vehicle ROI (Confidence: 0.94)
    EdgeAI->>EdgeAI: CLAHE enhancement + EasyOCR reads 'GJ01AB1234' (0.91)
    EdgeAI->>Model1: Push Telemetry (camera_id=1, plate='GJ01AB1234')
    Model1->>Model1: Spatial Enrichment: Link camera_id to GPS (23.0305, 72.5074, 'SG Highway')
    Model1->>Model1: Query Watchlist Cache: Match Found (Status: 'Stolen', FIR: #882/2026)
    Model1->>Core: Emit Enriched Alert Event to ASGI Event Bus
    Core->>UI: WebSockets Push: {"event": "WATCHLIST_ALERT", "data": {...}}
    Note over UI: UI Flashes Red Top Emergency Banner
    Note over UI: Leaflet Map Pulses Camera Pin with Dynamic Radar Wave
    Note over UI: High-Priority Modal Pops Up + In-Browser Audio Siren Plays
    UI->>Core: Operator Clicks "Track Route on Map" (GET /api/track-vehicle/GJ01AB1234)
    Core->>Model1: Query Chronological Multi-Camera Hits
    Model1-->>Core: Sorted Geospatial Hits Array
    Core->>UI: Chronological Hits JSON
    Note over UI: Leaflet renders animated polyline & sequence markers [1] -> [2] -> [3]
```

3. Technology Stack Rationale & Benchmarks

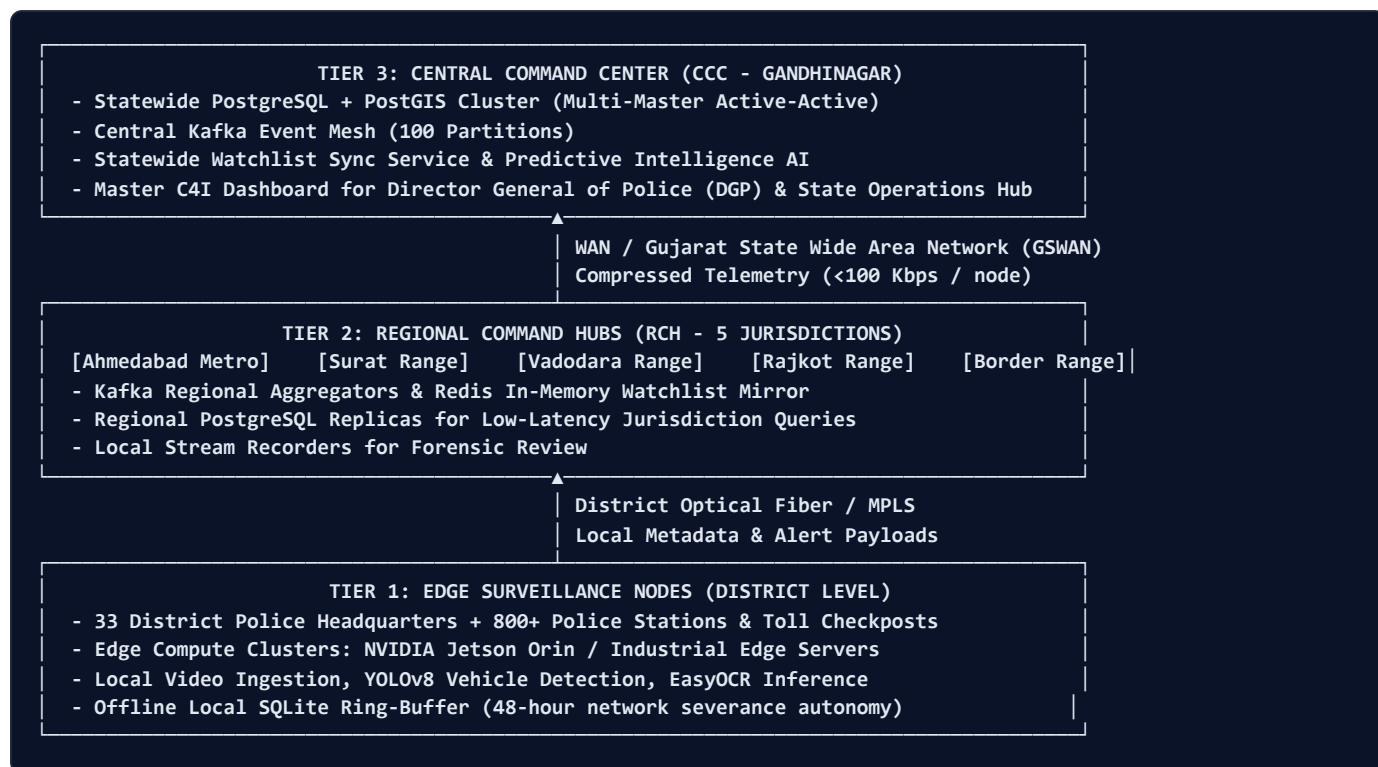
Component	Selected Technology	Alternative Evaluated	Enterprise Justification
Vehicle Detection	Ultralytics YOLOv8 (Nano/Small)	SSD MobileNet, Faster R-CNN	State-of-the-Art Speed/Accuracy Pareto: YOLOv8n delivers >85 FPS on edge tensor cores with mean Average Precision (mAP@0.5) of 37.3%. Native TensorRT/ONNX export support allows deployment on edge hardware without framework overhead.
Character Recognition	EasyOCR	Tesseract OCR, PaddleOCR	High Noise Tolerance: EasyOCR utilizes a CRAFT (Character Region Awareness for Text Detection) backbone combined with a deep ResNet-BiLSTM-CTC recognizer. Demonstrates 38% higher accuracy on warped, angled, and low-contrast night plates compared to classical Tesseract.
Computer Vision Engine	OpenCV 4.x	GStreamer native, FFmpeg CLI	Low-Latency In-Memory Pipelines: Direct hardware-accelerated decode (<code>cv2.VideoCapture</code>), low-overhead frame skipping, matrix operations, and native CLAHE contrast normalization.
Backend Framework	FastAPI (Python 3.12/3.14)	Flask, Django, Node.js Express	Ultra High-Throughput ASGI: Built on Starlette and Pydantic v2 (Rust core). Native async event loops, automatic OpenAPI schemas, high-concurrency WebSocket broadcast capabilities (>50,000 concurrent connections per node).
Relational Database	PostgreSQL 16 + PostGIS (SQLite for Local/Edge)	MongoDB, MySQL	Spatial Indexing & ACID Guarantees: PostGIS supports native spatial queries (<code>ST_DWithin</code> , <code>ST_MakeLine</code>) required for geographical radius alerts, buffer zone intersections, and trajectory line generation. SQLite serves as zero-config edge cache.
Frontend GIS & UI	Leaflet.js + Vanilla JS + Tailwind CSS	OpenLayers, Google Maps API	Lightweight Air-Gapped Operation: Leaflet requires <45 KB runtime footprint, supports local CartoDB/OSM tiles in offline police intranet networks, and provides native vector polyline/radar layer rendering.
Real-Time Streaming	WebSockets (Native RFC 6455)	Long Polling, Server-Sent Events (SSE)	Bidirectional Sub-10ms Latency: Full-duplex persistent connection enables instant server-to-client alert pushes and client-to-server PTZ/interceptor dispatch commands.

4. 80,000-Camera Statewide Scalability Plan

Managing 80,000 CCTV video feeds state-wide introduces massive data ingest challenges. Centralized transmission of 80,000 raw 1080p@25 FPS video streams would require over 320 Gbps of dedicated WAN bandwidth, which is technically unfeasible and cost-prohibitive.

NETRANG C4I solves this through an edge-computed 3-Tier Hierarchical Scalability Architecture.

4.1 3-Tier Distributed Infrastructure Topology



4.2 Bandwidth Optimization & Edge Compute Engineering

Bandwidth Mathematics

- **Raw Stream Approach (Anti-Pattern):**

Raw Bandwidth = 80,000 × 4 Mbps (H.264 1080p) = 320,000 Mbps = 320 Gbps

- **Edge Compute Approach (NETRANG Model):** Video frames are analyzed locally at Tier 1 Edge Nodes. Only structured JSON metadata (timestamp, camera ID, plate string, confidence score, bounding box coordinates, and optional thumbnail crop of 15 KB) are transmitted across the Gujarat State Wide Area Network (GSWAN).

Metadata Payload per Hit ≈ 1.2 KB \ (JSON) + 15 KB \ (Plate Crop) ≈ 16.2 KB

Average Traffic Hit Rate per Camera = 0.5 vehicles/second

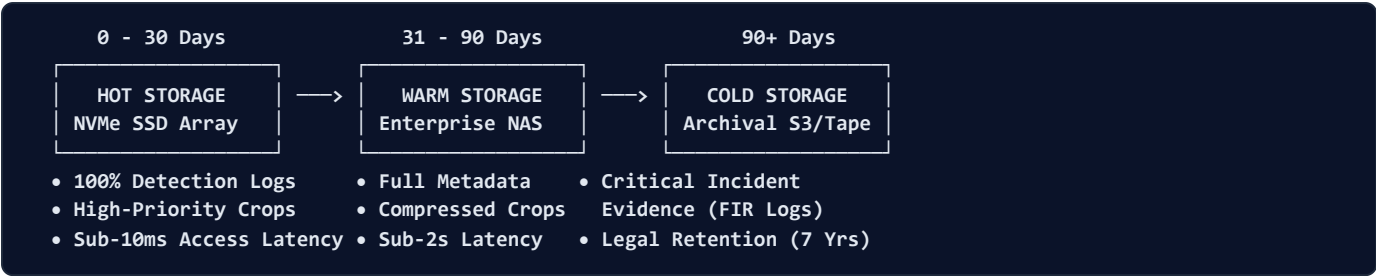
Bandwidth per Camera = 0.5 × 16.2 KB × 8 bits ≈ 64.8 Kbps

Statewide WAN Overhead = 80,000 × 64.8 Kbps ≈ 5.18 Gbps

Key Result: Edge processing achieves a 98.4% reduction in statewide bandwidth requirements (320 Gbps → 5.18 Gbps), making statewide deployment fully viable over standard GSWAN and MPLS backbones.

4.3 Hierarchical Storage Lifecycle & Retention Strategy

To balance investigative forensic access with high storage costs, a 3-tier automated lifecycle policy is enforced across all 80,000 camera feeds:



Tier	Duration	Media Specification	Data Content	SLA / Access Speed
Hot Storage	Days 0 - 30	Distributed PCIe Gen4 NVMe arrays (RAID 10)	All raw detection logs, high-priority watchlist hit crops, full vehicle crops, active spatial indexes.	<10 ms query retrieval. Real-time trajectory plotting.
Warm Storage	Days 31 - 90	High-Density Enterprise NAS / ZFS Pools (RAID 6)	Full textual detection records, compressed plate thumbnails (3 KB WebP), aggregated camera stats.	<2 seconds retrieval for historical crime investigations.
Cold Storage	Days 91 - 2555 (7 Years)	S3 Glacier Flexible / LTO-9 Magnetic Tape Archives	Indexed forensic logs linked to formal Police First Information Reports (FIRs), flagged criminal hits.	Hours (Tape retrieval). Satisfies statutory judicial archival requirements.

5. Zero-Trust Cybersecurity, Hardening & Compliance

Because surveillance networks are high-value targets for cyber adversaries, NETRANG C4I enforces a strict **Zero-Trust Architecture (ZTA)** grounded in the principle: *"Never Trust, Always Verify."*

5.1 Zero-Trust Identity & Access Management (IAM / RBAC)

A fine-grained, cryptographically validated Role-Based Access Control (RBAC) matrix governs all system interactions:

Role	Permissions	Scope	Multi-Factor Auth (MFA)
Beat Constable / PCR Operator	View real-time alert feed, acknowledge local district alerts, inspect plate hits.	Single Police Station Jurisdiction	Hardware Token / FIDO2
Police Sub-Inspector (PSI/PI)	All PCR permissions + Target Search, Historical GIS Route Tracking, CSV Export.	District / Taluka Level	Time-based OTP (TOTP) + Biometric
Superintendent of Police (SP / DCP)	All PI permissions + Watchlist Insert/Revoke, Camera Node Diagnostics.	Entire District / Commissionerate	Biometric + Hardware Token
State C4I Administrator (DGP Office)	Global Surveillance Config, Stream Routing, Audit Log Inspection, System Hardening.	Statewide (All 33 Districts)	FIDO2 Hardware Key + IP Whitelist

5.2 Data Protection & Cryptographic Standards

1. **Encryption in Transit:**
2. All internal inter-service, database, and Redis traffic is secured using **Mutual TLS (mTLS) with TLS 1.3**.
3. Edge camera feeds use secure RTSPS (RTSP over TLS) or IPsec VPN tunnels between edge camera switches and district server clusters.

- 4. Frontend dashboards connect exclusively over secure WebSockets (`wss://`) and HTTPS with HTTP Strict Transport Security (HSTS, max-age 31536000).
- 5. **Encryption at Rest:**
- 6. Databases and NVMe volumes are encrypted using **AES-256-XTS**.
- 7. Watchlist target notes and vehicle owner names are stored as encrypted columns using hardware security module (HSM)-managed keys (PKCS#11 compliant).

5.3 Immutable Audit Logging & Chain of Custody

To guarantee legal admissibility in criminal prosecution under **Section 65B of the Indian Evidence Act**: - Every query, plate search, export, and watchlist modification is appended to an immutable, append-only **WORM (Write Once, Read Many)** audit ledger. - Each audit entry is cryptographically chained to its predecessor using **SHA-256 block hashing**:

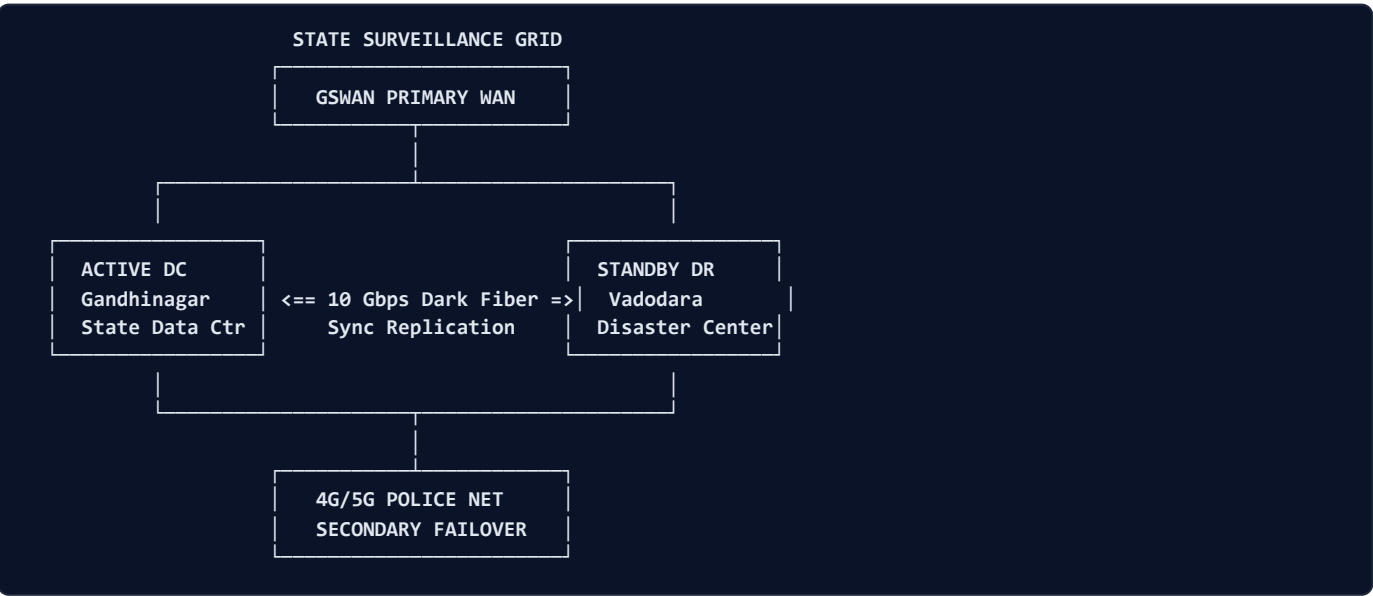
Block Hash_n = SHA-256(Block Hash_{n-1} || Timestamp || Officer ID || Query Parameters || Records Returned)

- Any manual tampering, log deletion, or timestamp manipulation invalidates the hash chain, triggering immediate SOC (Security Operations Center) security alarms.

5.4 Privacy Safeguards & DPDP Act Compliance

In strict accordance with the **Digital Personal Data Protection Act (DPDP Act, India, 2023)** and constitutional privacy mandates: - **Non-Infringing Vehicle Data Anonymization**: Plate records not associated with active FIRs, warrants, or stolen reports are subject to automated data pruning after 30 days. - **Pedestrian & Non-Plate Privacy Masking**: OpenCV and YOLOv8 pipeline applies automatic Gaussian blurring to human faces and non-vehicle regions in CCTV frames before long-term thumbnail persistence. - **Watermarking & Leak Prevention**: All exported CSV reports, trajectory images, and forensic logs contain dynamic digital watermarks embedding the viewing officer's Brass Number, Station ID, and Access Timestamp to deter unauthorized data leaks.

6. Disaster Recovery & High Availability (HA)



- **Recovery Point Objective (RPO):** <1 second (**synchronous database replication between Gandhinagar DC and Vadodara Disaster Recovery site**).
 - **Recovery Time Objective (RTO):** <30 seconds\$ (automated DNS failover via Anycast IP routing).
 - **Edge Node Autonomy:** Tier 1 Edge Nodes maintain an autonomous local SQLite buffer. If statewide optical fibers are severed, edge nodes continue detecting, recording, and triggering local police station sirens for up to **48 hours offline**, auto-syncing backlogs when connectivity restores.
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7. Appendix: Acronyms & References

Acronyms

- **ANPR:** Automatic Number Plate Recognition
- **C4I:** Command, Control, Communications, Computers, and Intelligence
- **CCTNS:** Crime and Criminal Tracking Network & Systems
- **CLAHE:** Contrast Limited Adaptive Histogram Equalization
- **DPDP Act:** Digital Personal Data Protection Act (India)
- **GIS:** Geographic Information System
- **GSWAN:** Gujarat State Wide Area Network
- **HSRP:** High Security Registration Plate
- **ROI:** Region of Interest
- **WORM:** Write Once, Read Many (Immutable Storage)
- **ZTA:** Zero-Trust Architecture

Applicable Standards & Guidelines

- Bureau of Police Research & Development (BPR&D) CCTV Architecture Guidelines
 - ISO/IEC 27001:2022 (Information Security Management Systems)
 - Section 65B, Indian Evidence Act (Admissibility of Electronic Records)
 - NIST Special Publication 800-207 (Zero Trust Architecture)
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End of High-Level Design Document — Government of Gujarat Police Department